ME 235: Enhanced Perfluorocarbon Delivery Systems for Emergency Medicine: Integrating Magnetic Clamshell and Deformable Mesh Technologies

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Abstract

With this project, we hope to better understand how perfluorocarbons (PFCs) can be used in emergency medicine to aid in maintaining blood pressure and tissue oxygenation. Perfluorocarbons are a class of materials composed of carbon and fluorine atoms which are both extremely stable and capable of containing high volumes of dissolved gases. Leveraging these properties poses PFCs as candidates for synthetic blood substitutes, which could aid in situations of severe blood loss. In this project, we would seek to determine the real-life effectiveness of using PFCs in clinical settings, analyzing their viscosities, dissolved gas storage capabilities, and gas diffusion properties to assess their similarities to natural human blood.

Building upon our foundational approach of PFC-encapsulated biconcave disks with magnetic clasp mechanisms, we propose an enhanced system: deformable PDMS mesh carriers that undergo controlled deformation in capillaries, enabling precise spatiotemporal release of PFC-gel for optimized oxygen transport. This approach, inspired by biological pressure-responsive systems such as starfish stomach eversion, addresses limitations in current synthetic blood substitute technologies by providing controlled release timing and bidirectional oxygen transport capabilities.

1 Project Significance

Improving the accessibility and simplicity of blood transfusions by developing a suitable synthetic blood substitute could represent a major step forward in medical technology. As described by the Mayo Clinic, blood transfusions are both common and crucial, but they rely on a compatible human blood supply to be performed (1). Additionally, according to the Red Cross, donated blood won't last forever, with the oxygenation-critical red blood cells expiring after only six weeks (2). Our background sources indicate that PFCs have been examined as potential synthetic blood substitutes (3; 4; 5), and a better understanding of their capabilities for tissue oxygenation could provide some insight into the challenges of the development of synthetic blood substitutes.

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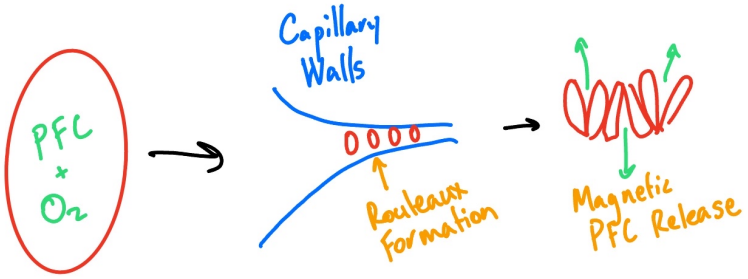

Figure 1: As synthesized biconcave disks enter capillaries, Rouleaux formation causes them to open, releasing oxygenated PFC. While effective, this may come with side effects of free PFC in the blood, and re-oxygenation of the PFC is quite difficult.

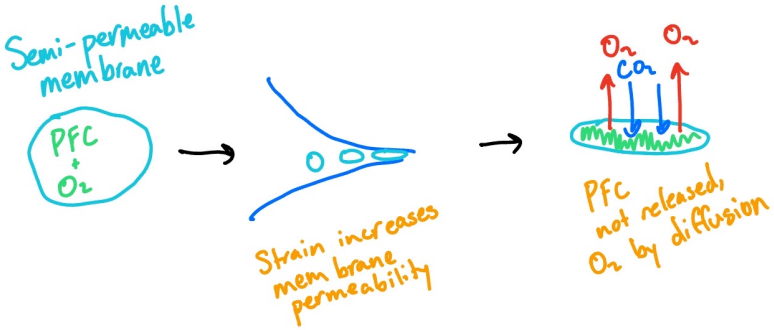

Figure 2: By leveraging strain-dependend permeability, we propose synthetic cells which can perform repeated gas exchange by diffusion, limiting the exposure of the blood to free PFCs

The proposed deformable mesh system represents a shift from passive to active, pressure-responsive oxygen delivery. Current PFC formulations rely primarily on bulk concentration gradients (6), but our integrated system provides targeted release where PFC-gel extrusion occurs only under specific pressure conditions in capillaries (Figure 2). In addition, our system offers reversible oxygen transport through extruded gel matrices, physiologically-integrated activation via natural blood pressure dynamics, and enhanced biocompatibility through controlled release that reduces systemic PFC exposure while maintaining therapeutic efficacy.

2 Methodology and Analysis

Our study will use a systematic approach to evaluate both magnetic clasp biconcave disks and deformable mesh carriers as artificial blood substitute technologies.

2.1 PFC-PDMS Biconcave Disk Synthesis

First, we will develop encapsulated perfluorocarbon PDMS-based magnetic biconcave disks that mimic the shape and approximate size of red blood cells (7-8 μm). These will be fabricated using microfluidic techniques, specifically flow-focusing junction microfluidics where we will employ a double-emulsion approach using a flow-focusing device with three concentric channels (7). The inner channel will contain the PFC liquid, the middle channel will carry a PDMS pre-polymer solution containing dispersed magnetic nanoparticles (Fe_3O_4), and the outer channel will contain an aqueous surfactant solution. This configuration will generate core-shell droplets. The resulting spherical droplets will pass through specially designed constriction chambers that create controlled deformation forces, inducing the characteristic biconcave morphology before polymer crosslinking occurs. The shaped droplets will be exposed to UV light to rapidly crosslink the PDMS shell while maintaining the biconcave morphology, and we will vary the concentration and distribution of magnetic nanoparticles

within the PDMS matrix to achieve optimal magnetic response while maintaining required mechanical properties.

2.2 Deformable Mesh Carrier Fabrication

The enhanced system features deformable PDMS mesh structures that undergo controlled deformation when passing through capillaries. PDMS demonstrates biocompatibility, flexibility, and chemical stability for biomedical microfluidic applications (8). We will create microporous PDMS structures with controlled pore sizes (200-500 nm) using sacrificial template methods, optimizing mesh geometry with spherical carriers of 8-10 μm outer diameter and wall thickness of 500-800 nm to ensure activation occurs in target microvascular regions. Electrospun PDMS mesh materials show enhanced biocompatibility and permeability compared to solid PDMS films (9).

The carriers will be loaded with high-viscosity PFC-gel containing perfluorooctyl bromide (PFOB) as the primary oxygen carrier, along with biocompatible thickening agents. Plasma-compatible gels using synthetic crosslinkers demonstrate biocompatibility and controlled release properties for biomedical applications (10). When carriers enter capillaries, axial compression induces radial expansion according to Poisson's ratio effects, where $\varepsilon_{\text{radial}} = -\nu \times \varepsilon_{\text{axial}}$ with $\nu \approx 0.45 - 0.49$ for PDMS, creating the pressure threshold for gel extrusion.

2.3 Biotransport Principles and Analysis

The primary principles of this project involve studying the concentrations of oxygen and carbon dioxide in PFCs, along with their flow rates, pressure drops, and diffusion coefficients in a biological context. This way, we can determine how PFCs might behave when driven by a human heart through blood vessels. The threshold pressure for gel extrusion follows

$$P_{\text{threshold}} = \frac{4\gamma \cos\theta}{d_{\text{pore}}} + \sigma_{\text{yield}} \tag{1}$$

where γ represents surface tension of PFC-gel ($\sim$15-20 mN/m), θ is the contact angle with PDMS ($\sim 110°$), d_{pore} is the mesh pore diameter, and σ_{yield} is the yield stress of the gel matrix.

Flow dynamics in our system are governed by modified Poiseuille flow accounting for deformable carriers

$$\Delta P_{\text{total}} = \Delta P_{\text{fluid}} + \Delta P_{\text{deformation}} + \Delta P_{\text{extrusion}} \tag{2}$$

with target operating range of $\Delta P = 20 - 40$ mmHg representing physiological capillary pressure. Oxygen transport kinetics through the extruded gel matrix follow

$$\frac{\partial C}{\partial t} = D_{\text{eff}} \times \nabla^2 C - k_{\text{release}} \times C \tag{3}$$

where D_{eff} accounts for tortuosity in the gel network: $D_{\text{eff}} = D_{\text{free}} \times (\phi/\tau)$ with ϕ representing porosity of gel matrix (0.6-0.8) and τ as the tortuosity factor (1.5-2.5).

2.4 Oxygen Transport Characterization

We will measure the oxygen carrying capacity and release kinetics of our PFC-based discs compared to natural red blood cells using oxygen dissolution measurements at various partial pressures, oxygen release rates under simulated tissue conditions, and comparison of oxygen dissociation curves between PFC discs and blood. Flow dynamics assessment will include measuring the viscosity of PFC solutions at various concentrations, analyzing flow behavior in microfluidic channels simulating capillaries, comparing pressure drops across microchannel networks, and evaluating deformation characteristics during passage through narrow channels.

2.5 Diffusion Studies

We will quantify oxygen diffusion rates from PFC disks to surrounding tissue-mimicking matrices, compare oxygen concentration gradients between PFC systems and blood, and measure time to equilibration in tissue-mimicking environments. Mathematical modeling will employ finite element analysis using COMSOL simulations of mesh deformation under flow conditions, coupled fluid-structure interaction modeling, and optimization of mesh geometry and material properties alongside mass transfer modeling using reaction-diffusion equations for oxygen transport in tissue.

3 Preliminary Experimentation: Computational Modeling and Simulation

To evaluate the effectiveness of our proposed PFC delivery systems, we developed a MATLAB simulation modeling oxygen transport in an artery-capillary system. The simulation compares three scenarios: traditional red blood cells (RBCs), free PFC emulsions, and our proposed encapsulated PFC system with controlled release.

3.1 Simulation Setup and Geometry

The computational model represents a coupled artery-capillary segment with physiological dimensions. The artery segment spans 500 μm in length with a 50 μm radius, transitioning to a 200 μm capillary segment with a 4 μm radius, surrounded by tissue extending to 50 μm radius. This geometry captures the transition zone where oxygen transport mechanisms differ between arterial and capillary environments.

The simulation employs a finite difference approach with grid resolution (101 points in the artery, 81 in the capillary) and incorporates flow velocities of 5000 μm/s in arteries and 1000 μm/s in capillaries. Physical parameters include diffusion coefficients for oxygen in blood (3000 μm^2/s), tissue (2000 μm^2/s), and arterial walls (1000 μm^2/s), with tissue consumption following Michaelis-Menten kinetics characterized by maximum consumption rate $M_{\text{max}} = 20.0$ μM/s and Michaelis-Menten constant $K_m = 5.0$ μM.

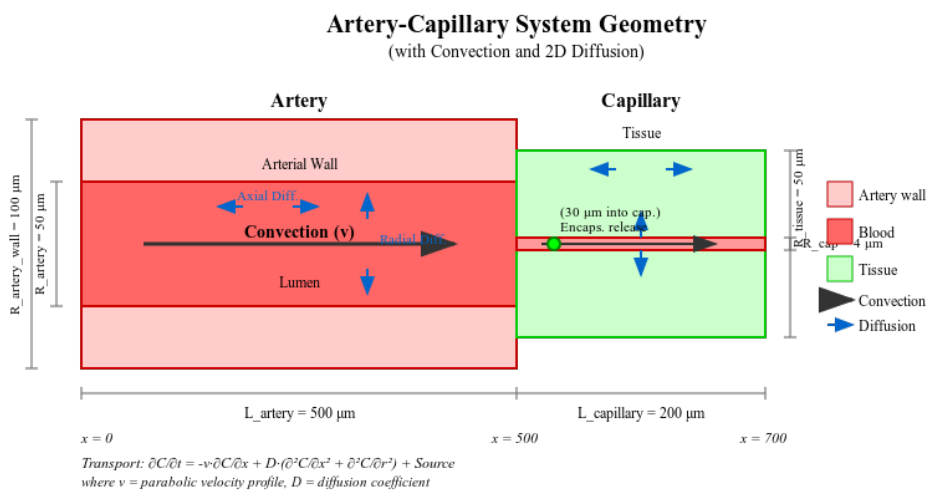

Figure 3: Rough diagram of the geometry used in the MATLAB simulation

3.2 Carrier-Specific Release Mechanisms

For RBCs, the simulation implements cooperative oxygen binding following the Hill equation,

$$S_{\text{eq}} = \frac{C^{n_{\text{hill}}}}{P_{50}^{n_{\text{hill}}} + C^{n_{\text{hill}}}} \tag{4}$$

with Hill coefficient $n_{\text{hill}} = 2.7$ and half-saturation pressure $P_{50} = 80$ μM, coupled with release rates of 0.01 s^{-1} in arteries and 3.0 s^{-1} in capillaries to reflect physiological oxygen-hemoglobin dissociation behavior.

Free PFC systems follow Henry's law for linear oxygen solubility with release rates of 1.0 s^{-1} in both arterial and capillary segments, representing immediate equilibration with local oxygen partial pressure. Our encapsulated PFC system incorporates controlled release activation at a specific

location 30 μm into the capillary segment, where rapid release (12.0 s^{-1}) mimics the proposed pressure-activated mesh deformation mechanism.

3.3 Simulation Results and Performance Metrics

The simulation tracked key performance indicators including oxygen waste in arteries, effective delivery in capillaries, tissue oxygenation levels, and overall transport efficiency over 1.0 second simulation time. Results demonstrate differences between delivery mechanisms, with our encapsulated PFC system achieving zero oxygen waste in arteries (0.0% compared to 0.8% for RBCs and 90.2% for free PFC), substantial capillary delivery (1,424,248 μM total), and superior tissue oxygenation (226.0 μM compared to 82.9 μM for RBCs and 93.8 μM for free PFC).

The efficiency ranking places RBCs first (overall score = 0.253) due to their physiological optimization, our encapsulated PFC system second (score = 0.159) with the advantage of zero arterial waste, and free PFC systems last (score = 0.038) primarily due to oxygen waste in arterial segments. These results validate the concept of controlled release timing, demonstrating that strategic activation location can improve oxygen delivery efficiency while minimizing systemic waste.

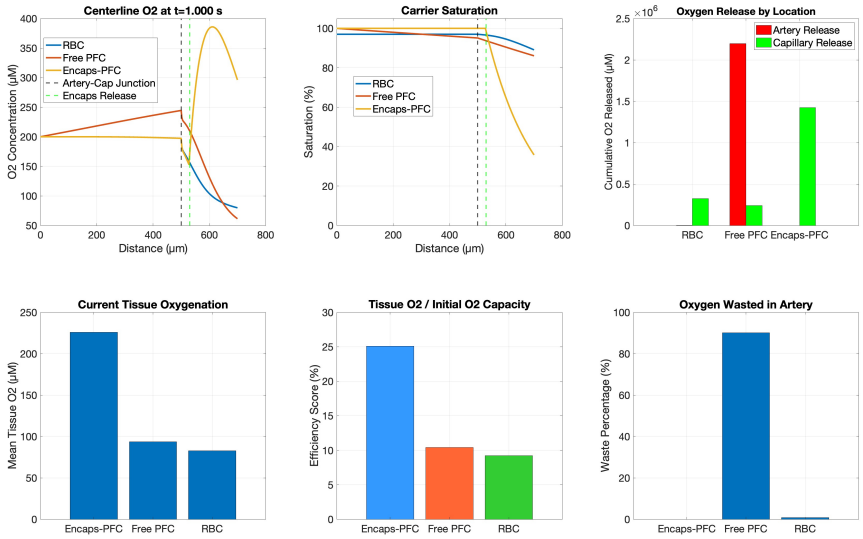

Figure 4: Computational comparison of oxygen transport mechanisms in artery-capillary system. (A) Centerline oxygen concentration profiles showing dissolved O_2 transport along 500 μm artery segment transitioning to 200 μm capillary segment for RBCs (blue), free PFC (orange), and encapsulated PFC (green). Vertical dashed lines indicate artery-capillary junction (black) and encapsulated PFC release location (green). (B) Carrier saturation profiles demonstrating oxygen binding capacity utilization along transport pathway. (C) Cumulative oxygen release by anatomical location comparing wasteful arterial release (red bars) versus useful capillary delivery (green bars). (D) Final tissue oxygenation levels achieved by each carrier system. (E) Transport efficiency scores calculated as ratio of tissue oxygenation to initial oxygen capacity. (F) Waste percentage showing fraction of total oxygen released prematurely in arterial segments. Encapsulated PFC system demonstrates zero arterial waste with substantial tissue oxygenation (226.0 μM), validating controlled release concept for synthetic blood substitute applications.

4 Expected Results

Our results, if successful, will highlight several key advances in synthetic blood substitute technology. The encapsulated PFC disks will have the same biconcave morphology of RBCs, thus will deform in a

controlled manner in the microchannels. Since the PFC disks have embedded magnetic nanoparticles, they will respond to external magnetic fields, enabling targeted delivery and controlled disintegration at the capillary length scale for optimized oxygen transport.

The deformable mesh carriers will demonstrate pressure-activated release with activation pressure thresholds of 25 ± 5 mmHg within physiological ranges, gel extrusion volumes of 60-80% of initial load under target conditions, and reversible deformation with minimal permanent set after repeated cycling. Oxygen transport efficacy will show initial release rates 2-3$\times$ faster than passive PFC emulsions, sustained release duration of 30-60 minutes in tissue-mimicking environments, and re-oxygenation efficiency exceeding 70

The PFC disks will release oxygen in simulated tissue conditions in a manner similar to RBCs. However, real blood will exhibit a sigmoidal oxygen dissociation curve, whereas PFCs may present linear profiles, which is consistent with prior literature. The concentration of PFCs can be tuned so that its viscosity and fluid flow properties match real blood. We anticipate that increasing the PFC concentration may increase the viscosity characteristics leading to higher pressure drops in the microchannel networks.

Flow characteristics will demonstrate minimal impact on bulk viscosity (increase $<$20% at therapeutic concentrations), smooth passage through 5-8 μm channels after activation, and controlled aggregation prevention via magnetic field manipulation. Clinical relevance will be established through oxygen delivery capacity equivalent to 20-30% hematocrit, reduced systemic PFC burden compared to conventional emulsions, and enhanced oxygen extraction efficiency in hypoxic tissue regions.

Our diffusion studies will show that both the biconcave PFC disks and deformable mesh carriers will achieve efficient oxygen transport, demonstrating their suitability as synthetic blood substitutes for clinical applications. The integration of magnetic clasp mechanisms with pressure-responsive mesh technology creates a multifunctional platform that mimics RBC oxygen transport while offering advantages in storage stability, compatibility, and targeted delivery, positioning this technology for clinical translation in emergency medicine applications.

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